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# **Wastewater Treatment Plant Design**

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## **EXECUTIVE SUMMARY**

Executive summary to be added here afterwards

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# 1 INTRODUCTION AND CONCEPTUAL DESIGN DESCRIPTION

Population and economical growth among other factors require the subsequently development of cities and the general infrastructure that supports them. As a cities expand, the electrical grid, road systems, drinking water and Wastewater recolection and treatment, among others, need to increase their capabilities as well. In the recent years, a northwest area in Melbourne has been rapidly developping and therefore, requires the construction of a new Wastewater treatment plant that can address the current and future needs of the community. In the presente report, the authors present the design of the abovementioned plant following local guidlines.

The designed is conformed by a *Preliminary Treatment - Section 2* where the screening is designed to remove coarse material on the flow stream to protect downstream process equipment and increse its reliability [Metcalf and Eddy, 2014]. The next step is the *Primary Sedimentation - Section 3*, designed to remove the settled solids and floating material. This process can reach removals ratios of 50% to 70% of the total suspended solids (TSS) and 25% to 40% of the BOD. It is followed by the *Secondary Treatment - Section 4*, in which sludge and activation systems are used to remove the soluble, colloidal and particulate suspended organic substances by coagulating them through the microorganisms (sludge).

Finally, the environmental impact of the incorporation of the treated wastewater into the waterway is assessed against the local guidelines applicable to determine whether it complies with its requirements or if further treatment shall be implemented.

## 1.1 Population estimations

To determine the waswater flowrates and mass loads for develoment areas that the plant should be capable of process, where actual flow and load measurements are not avilable, the population equivalent can be used. This is a standarization of the wastewater production and characteristics based on the amount of people that would produce the same values. For example, in the *AMF L8 Sewerage Planning and Design Principles* which is the standard for wastewater treatment plants, developed by *Yarra Valley Water* which covers the area underdesign in the present report, on its *Table 3-4* equivalent population factors are presented for different development types for both residential and commercial use. For this report, based on previous surveys and analysis, it is known that the equivalent population from domestic contributions ( $PE_{\text{domestic}}$ ) is equal to 340000. Similarly, for commercial contributions, the equivalent population is known to comply with the following calculation:  $PE_{\text{commercial}} = \lceil \sqrt{10} \times 9000 \rceil = 28461$ .

## 1.2 Current flowrates estimation

According to the *Water Service Association of Australia* the inflow rate percapita can be estimated as 180 L/PE/day [Yarra Valley Water, 2023]. Therefore, the  $Q_{\text{current avg, domestic}}$  can be calculated as

$\frac{Q_{\text{domestic per capita}} \times PE_{\text{domestic}}}{1000} = 61200.0 \text{ m}^3/\text{day}$ . Similarly, based on previous surveys it got determine that the commercial contribution for this area is 280 L/PE/day. Therefore the  $Q_{\text{current avg, commercial}}$  can be calculated as  $\frac{Q_{\text{commercial per capita}} \times PE_{\text{commercial}}}{1000} = 7969.08 \text{ m}^3/\text{day}$ .

The infiltration/inflow flowrate, when no data is available, can be estimated based on either the total length of the sewer system, or based on the area to be covered. For the last one, the ratio can

vary from 0.2 up to 28 m<sup>3</sup>/ha/day [Metcalf and Eddy, 2014]. In this report, the authors has choosen the lower limit as the system will be new and therefore the damage on the pipelines will be minimum.

As no data of the covered area is available, the autors have estimated it based on the average population per area ratio of Melbourne suburbs. The Melbourne population density per suburb data is available in the *Australian Bureau of Statistics* web page, see Apendix 1 for the raw data. From it, the Population average per suburb ( $P_{average}$ ) and the average landscape area ( $A_{average}$  km<sup>2</sup>) were calculated. With the  $P_{average} / A_{average}$  ratio, multiplied by the  $PE_{domestic}$  the estimated area of the understudy suburb ( $A_{estimated}$ ) is obtained. The  $PE_{commercial}$  is not included as it does not represents actual population. In the Table 1 the obtained results for the infiltration rate estimation. In the Table 2 the current flow rates obtained are summarized.

Table 1: Understudy landscape area estimation

| $P_{average}$ | $A_{average}$ (km <sup>2</sup> ) | $A_{average} / P_{average}$ | $A_{estimated}$ (km <sup>2</sup> ) | Ratio (m <sup>3</sup> /ha/day) | $Q_{infiltration}$ (m <sup>3</sup> /day) |
|---------------|----------------------------------|-----------------------------|------------------------------------|--------------------------------|------------------------------------------|
| 11200.0       | 18.65                            | 0.0017                      | 566.17                             | 0.2                            | 11323.41                                 |

### 1.3 Minimum current flow rates

The minimum flow rates can be estimated using the minimum peaking factor which from the 1 is equal to 0.3. The current flow rates shown in the Table 2 are multiplied by this factor, and results are presented in Table 3.

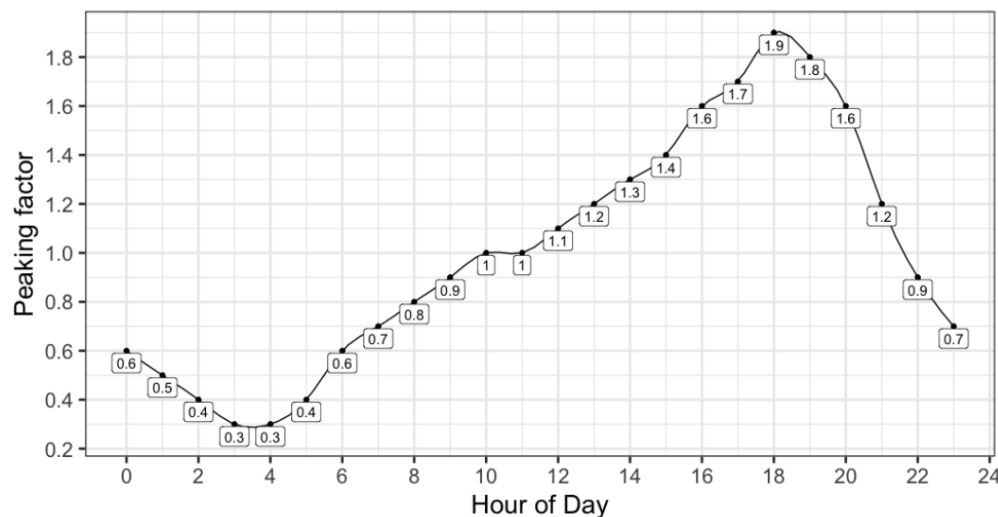


Figure 1: Daily fluctuation of peaking factors at each hour of the day for flow rates on an average day in a similar wastewater catchment.

### 1.4 Maximum current flow rates

The maximum flow rates can be estimated using the maximum peaking factor which from the Figure 1 is equal to 1.9. The current flow rates shown in the Table 2 are multiplied by this factor, and results are

presented in Table 3.

## 1.5 Population Growth and Future Design Considerations

To account for future demand on the wastewater treatment plant, the design incorporates a projected population growth. Based on previous analysis, it was determined that the annual growth reate can be estimated as:  $G_{rate} = 1 + 10^{0.015} = 2.04\%$ . The current population is therefore projected during the life span of the plant. The life span is commonly adopted as 20 years as it balances the need to consider future growth and technological advancements while ensuring economic feasibility [Water Environment Federation, 2017]. The projected population can be calculated based on (Eq. 1).

$$PE_{\text{projected}} = \left[ PE \times \left( 1 + \frac{\text{Growth Percentage}}{100} \right)^{20} \right] \quad (\text{Eq. 1})$$

With the per capita flow rates mentioned in the Section 1.2, following the same procedure there mentioned the flow rates for the projected population can be calculated as well. Additionally, the minimum and maximum projected flowrates can be calculated following the peak minimum factor and peak maximum factor mentioned in Sections 1.3 and 1.4, respectively. Results are shown in the Table 3.

To project the infiltration, a different approach is used. This value is not controlled by the population as it depends more on the environmental phenomenon. Because of this, it is projected based on an environmental factor instead. The *Environment, Land, Water and Planning* department of the *Victoria State Government* in their *Guidelines for the Adaptive Management of Wastewater Systems Under Climate Change in Victoria*, especified that the climate change factor can be taken as 1.2. The infiltration flow rate presented in Table 2 is therefore multiplied by this factor and presented in the Table 3.

Table 2: Current population and flow rates

| Contribution type | Population Equivalent | Minimum flow rate (m <sup>3</sup> /day) | Average flow rate (m <sup>3</sup> /day) | Maximum flow rate (m <sup>3</sup> /day) |
|-------------------|-----------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| Domestic          | 340000                | 18360.00                                | 61200.00                                | 116280.00                               |
| Commercial        | 28461                 | 2390.72                                 | 7969.08                                 | 15141.25                                |
| Infiltration      | —                     | 11657.00                                | 11657.00                                | 11657.00                                |
| Total             | —                     | 32407.72                                | 80826.08                                | 143078.25                               |

Table 3: Projected population and flow rates

| Contribution type | Population | Minimum flow rate (m <sup>3</sup> /day) | Average flow rate (m <sup>3</sup> /day) | Maximum flow rate (m <sup>3</sup> /day) |
|-------------------|------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| Domestic          | 427363     | 23077.60                                | 76925.34                                | 23077.60                                |
| Commercial        | 35775      | 3005.10                                 | 10017.00                                | 3005.10                                 |
| Infiltration      | —          | 13988.40                                | 13988.40                                | 13988.40                                |
| Total             | —          | 40071.10                                | 100930.74                               | 179178.85                               |

## 1.6 Influent Pollutant Load Estimation

Following the guidelines established for wastewater treatment design [Water Environment Federation, 2017, Metcalf and Eddy, 2014], the analysis focused on two key pollutants: Biochemical Oxygen Demand (*BOD*) and Total Suspended Solids (TSS), which are considered the most relevant indicators for organic and solids loading.

As no specific influent quality data was available for the study area, representative pollutant concentrations were obtained from Melbourne Water's open dataset on the Eastern Treatment Plant (ETP) [Melbourne Water, 2024]. This dataset, which includes daily influent quality records from 2014 onwards, is publicly accessible at <https://data-melbournewater.opendata.arcgis.com>. In Figure 2, the box plot of the (*BOD*) of the abovementioned data is presented.

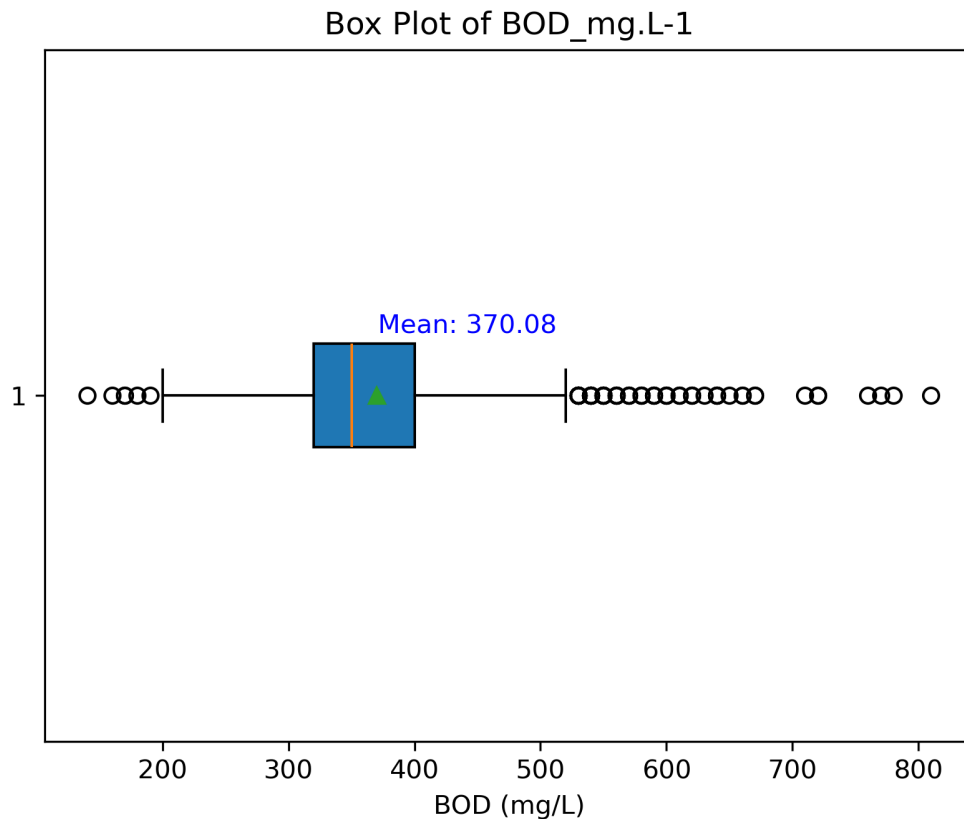


Figure 2: Boxplots of Biochemical Oxygen Demand (BOD) from Melbourne Water ETP dataset

## 2 PRELIMINARY TREATMENT DESIGN

### 2.1 Facility types

Coarse screens designed to retain gross pollutants greater than 10 mm are proposed to effectively protect downstream treatment processes. These screens, commonly employed in wastewater treatment plants

(WWTPs), efficiently remove large solids, rags, and debris (Davis, 2010). However, it is noted that their contribution to the removal of biochemical oxygen demand (BOD) is considered negligible.

Each mechanical screen unit should be installed in its own dedicated channel, facilitating isolation for maintenance or emergency purposes. Additionally, vortex grit chambers are proposed to remove grit particles up to 100 mm. This inclusion significantly reduces grit accumulation in downstream units and prevents sediment buildup within the equalisation basin, thus enhancing the overall reliability and effectiveness of subsequent treatment processes.

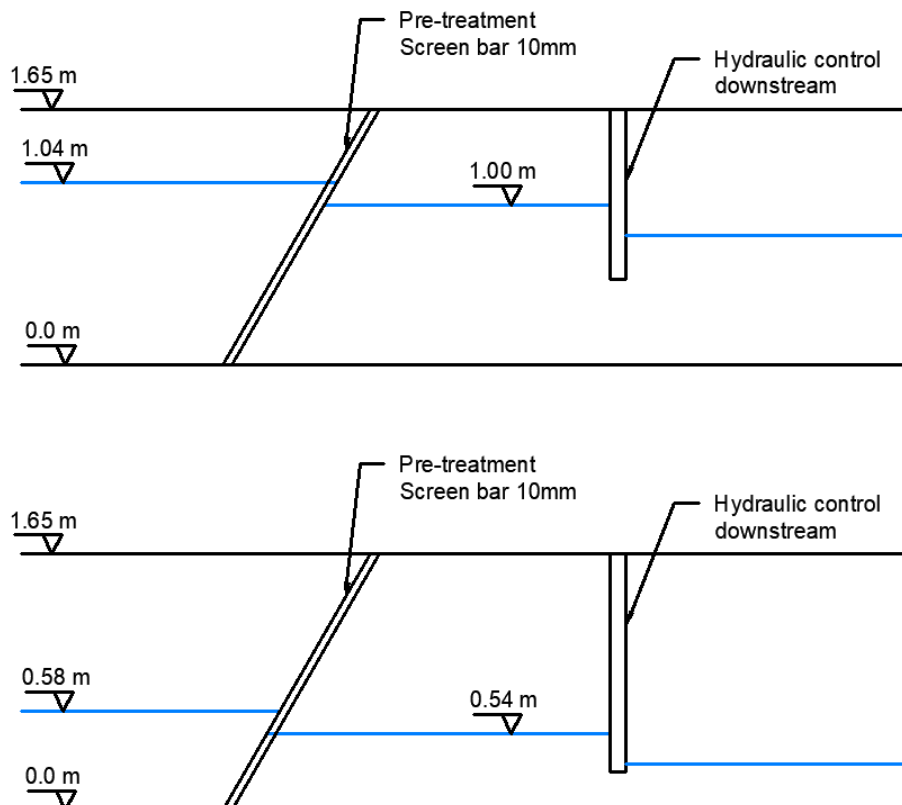


Figure 3: Preliminary treatment, screens section.

## 2.2 Screens design

When the flow goes through the screens, it might suffer a headloss due to the obstruction and turbulence created by the screens limited open area. This headloss can be neglected in the primary screening as the opening area remains big enough to not cause significant headloss. However, in the case of secondary screens, as the open area is smaller, the headloss needs to be accounted for. To determine its value and sizing of the secondary screening, the velocity upstream of the bars, the hydraulic head loss, and the cross-sectional flow area are calculated. The design assumes that flow depth is governed by the downstream hydraulic condition, and the channel width is then determined accordingly. The calculations are based on the peak projected flow rate.

The values used for this calculation are summarised in Table 4. The flow velocity through the screen  $v_{\text{screen}}$  should be between 0.4 and 0.9 m/s ([Davis, 2010]), the latter is selected and checked against the other constraints of the design. The bar spacing or opening size ( $b$ ) is taken as 0.01 m as the screen

should remove particles bigger than 10mm. The bar width  $w$  is selected as 0.007 m based on commercial screens availability. The discharge coefficient ( $C_d$ ) is commonly taken as 0.84, although it might be adjusted based on the screening specifications. Finally, the minimum required downstream flow depth was determined to be 1.0 m based on previous analysis.

Table 4: Design input values for screen and channel

| Parameter           | Value                 | Description                      |
|---------------------|-----------------------|----------------------------------|
| $v_{\text{screen}}$ | 0.9 m/s               | Flow velocity through the screen |
| $b$                 | 0.01 m                | Spacing between bars             |
| $w$                 | 0.007 m               | Bar width                        |
| $C_d$               | 0.84                  | Discharge coefficient            |
| $g$                 | 9.81 m/s <sup>2</sup> | Gravitational acceleration       |
| $d$                 | 1.0 m                 | Assumed downstream flow depth    |

From Equations (2) to (5) the upstream velocity ( $v_{\text{upstream}}$ ), the headloss ( $h_{\text{loss}}$ ), the upstream flow depth ( $d_{\text{upstream}}$ ), the wetted cross-sectional area ( $A_{\text{wet}}$ ) and required channel width ( $W_{\text{channel}}$ ) were determined. The screens width is recommended to be between 0.6 and 1.1m [Metcalf and Eddy, 2014]. The number of screens required ( $N_{\text{screens}}$ ) is therefore calculated based on the channel width divided by the maximum recommended screen width (1.1 m) and rounding up to the nearest integer. Results are shown in the Table 5.

$$v_{\text{upstream}} = v_{\text{screen}} \cdot \frac{b}{b + w} \quad (2)$$

$$h_{\text{loss}} = \frac{1}{2gC_d^2} (v_{\text{screen}}^2 - v_{\text{upstream}}^2) \quad (3)$$

$$A_{\text{wet}} = \frac{Q_{\text{peak}}}{v_{\text{upstream}}} \quad (4)$$

$$W_{\text{channel}} = \frac{A_{\text{wet}}}{d + h_{\text{loss}}} \quad (5)$$

$$W_{\text{screen}} = \frac{W_{\text{channel}}}{N_{\text{screens}}} \quad (6)$$

Table 5: Screen and channel design results

| Output                | Value              | Description                   |
|-----------------------|--------------------|-------------------------------|
| $v_{\text{upstream}}$ | 0.506 m/s          | Velocity before the screen    |
| $h_{\text{loss}}$     | 0.04 m             | Head loss across the screen   |
| $d_{\text{upstream}}$ | 1.04 m             | Flow depth upstream           |
| $A_{\text{wet}}$      | 4.1 m <sup>2</sup> | Wetted cross-sectional area   |
| $W_{\text{channel}}$  | 4.0 m              | Required screen channel width |
| $N_{\text{screens}}$  | 4                  | Number of screens required    |
| $W_{\text{screen}}$   | 1.0 m              | Screen width                  |

### 2.3 Blockage Scenario

To ensure robustness of the preliminary screening system, the design is evaluated under a conservative condition of 50% screen blockage. This is important as screen clogging can lead to increased velocities and head losses, and subsequent failure of the system.

The analysis is based on the Bernoulli equation applied between the upstream and downstream sections of the screen channel:

$$h_1 + \frac{v_1^2}{2g} = h_2 + \frac{v_2^2}{2g} + \Delta h \quad (7)$$

where: -  $h_1$  and  $h_2$ : upstream and downstream water depths, respectively [m] -  $v_1$  and  $v_2$ : corresponding flow velocities [m/s] -  $\Delta h$ : head loss due to the screen [m] -  $g$ : gravitational acceleration (9.81 m/s<sup>2</sup>).

The upstream water depth  $h_1$  is determined by:

$$h_1 = \frac{h_2 \cdot v_2}{v_1} \quad (8)$$

The head loss across a partially blocked screen is calculated using a modified version of Eq.(3) to account for the blockage ratio which affects the discharge velocity coefficient ( $v_{\text{coef}}$ ):

$$\Delta h = \frac{(v_{\text{coef}}^2 - 1^2) \cdot v_1^2}{2gC_d^2} \quad (9)$$

$$v_{\text{sc}} = \frac{Q/N_{\text{screens}}}{h_2 \cdot \left( \frac{W_{\text{screen}} - w}{w + b} \cdot w \right)} \quad (10)$$

$$v_2 = \frac{Q/N_{\text{screens}}}{h_2 \cdot W_{\text{screen}}} \quad (11)$$

$$h_1 = \frac{h_2 \cdot v_2}{v_1} \quad (12)$$

$$v_{\text{coef}} = \frac{w + b}{0.5 \cdot w} \quad (13)$$

Substituting Eqs. (8) and (9) into the Bernoulli equation (7), it is obtained:

$$\frac{h_2 v_2}{v_1} + \frac{v_1^2}{2g} = h_2 + \frac{v_2^2}{2g} + \frac{(v_{\text{coef}}^2 - 1^2) \cdot v_1^2}{2gC_d^2} \quad (14)$$

Solving Eq.(14) for  $v_1$  leads to a third grade equation, which is solved for both the current average and projected peak flowrates, see Tables 2 and 3. The purpose is to ensure that even with 50% blockage, the upstream velocity remains within acceptable hydraulic limits (typically not exceeding 1 m/s) and the head loss does not cause excessive backwater.

The obtained velocities are summarized in Table 6 and 7 for current average and projected peak condition, respectively. It can be noticed that for both cases, the velocities remain within the acceptable values.

*Table 6: Blockage scenario - current average flow*

| Parameter | Result | Units | Description            |
|-----------|--------|-------|------------------------|
| $v_{sc}$  | 0.97   | m/s   | Screen design velocity |
| $v_2$     | 0.54   | m/s   | Velocity downstream    |
| $v_1$     | 0.41   | m/s   | Velocity upstream      |

*Table 7: Blockage scenario - projected peak flow*

| Parameter | Result | Units | Description            |
|-----------|--------|-------|------------------------|
| $v_{sc}$  | 0.89   | m/s   | Screen design velocity |
| $v_2$     | 0.52   | m/s   | Velocity downstream    |
| $v_1$     | 0.44   | m/s   | Velocity upstream      |

### 2.3.1 Operational considerations

During the screens operation, a target velocity of  $0.7 - 0.9 \text{ m/s}$  at peak flows is ideal for operational balance, ensuring effective screenings removal without excessive turbulence or debris bypass. The channel dimensions were sized to maintain appropriate flow velocities under varying conditions and considering future growth and possible peak events. Finally, the screens should be easily accessible and designed for rapid maintenance response. Scheduled inspections and routine cleaning are required, especially after high flow events.

## 2.4 Equalisation basin

The equalisation basin was designed to minimise flow fluctuations to downstream treatment units, enhancing overall plant stability and performance. Initially, the basin volume calculated by capturing all inflows exceeding the projected average daily flow resulted in an impractical tank size ( $24,000 \text{ m}^3$ ). Therefore, a new peak flow of  $138,626 \text{ m}^3/\text{day}$  was established, based on the maximum capacity of the downstream primary treatment facilities. (See Section 3). This revised approach operates near-average flow rates for most hours of the day. During peak inflow periods (from 3 pm to 9 pm), inflows exceeding this new peak flow rate are diverted to the equalisation basin. In low-demand hours, stored wastewater is returned to maintain less variable flow to the downstream processes. This method reduced the required equalisation volume and significantly reduced the peak flows. After recalculating using this approach, the required equalisation volume was smaller, although this requires extra operational control, providing narrower variation in the hydraulic regime for downstream processes. This equalisation basin requires aeration so as to prevent odours and solids deposition. The proposed equalisation is an off-line basin as seen in the diagram shown in Figure 4.

## 2.5 Protection against failure

To ensure continuous operation and protect against failure scenarios, the preliminary treatment facilities incorporate redundancy and safety features, described in Davis, 2010 and summarized in Table 8

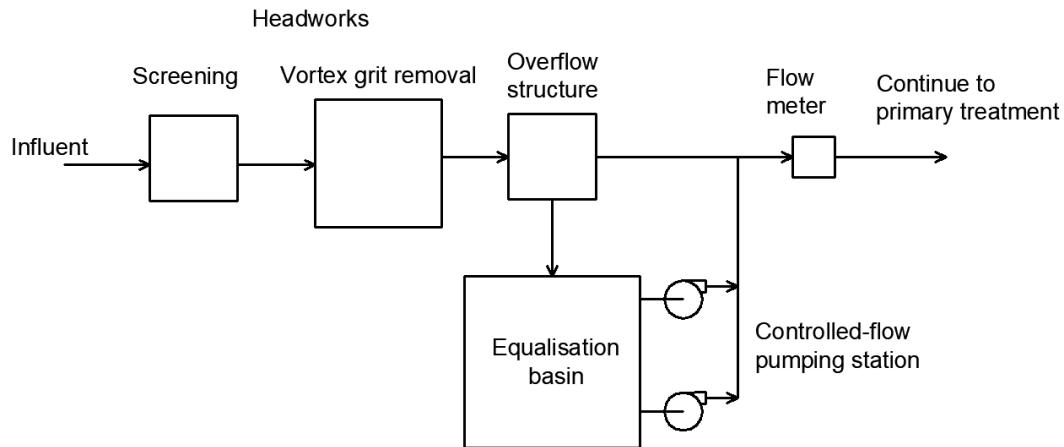


Figure 4: Off-line equalisation basin –adapted from Fig 3-20, Davis,2010.

### 3 PRIMARY TREATMENT

#### 3.1 Facility Types

Primary treatment in this design is achieved through the use of circular primary clarifiers. While both circular and rectangular clarifiers are widely used, circular tanks were selected due to their hydraulic efficiency, ease of sludge removal, and suitability for continuous operation in medium-to-large wastewater treatment plants. Circular clarifiers promote uniform radial flow distribution, reducing dead zones and short-circuiting, which enhances sedimentation performance. Mechanically, the rotating sludge scraper arms in circular clarifiers allow for reliable and consistent solids removal with minimal manual intervention. Moreover, for a greenfield development with expected long-term service and increasing population, circular clarifiers provide a scalable, space-efficient, and operationally robust solution.

#### 3.2 Primary Sedimentation

##### 3.2.1 Clarifier Sizing and Flow Equalization

To determine the volume of the primary clarifiers, the removal efficiencies required for suspended solids (TSS) and biochemical oxygen demand (BOD) were referenced from standard design tables [Metcalf and Eddy, 2014]. The detention times required for each are calculated using Eq.15 in conjunction with Eq.16. This last one is to increase the retention time when the temperature is lower than 20 degrees as the water viscosity gets affected. For this report, a temperature  $T$  of 10 degrees was chosen to incorporate this phenomenon, although further analysis could lead to adjust this value.

$$t = \left( \frac{E_r \cdot a}{1 - E_r \cdot b} \right) \cdot T_{\text{factor}} \quad (15)$$

$$T_{\text{factor}} = \begin{cases} 1.82 \cdot e^{-0.03 \cdot T}, & \text{if } T < 20^\circ\text{C} \\ 1, & \text{if } T \geq 20^\circ\text{C} \end{cases} \quad (16)$$

Table 8: Operational resilience and contingency measures [Davis, 2010]

| Aspect                                 | Description                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Redundancy Measures</b>             | An additional mechanically cleaned coarse screen and vortex grit chamber unit will be provided as backup units. This ensures operational continuity during periods when primary units are undergoing maintenance or experiencing overloading conditions. |
| <b>Bypass Channels and Overflow</b>    | Dedicated bypass channels will be installed, allowing flows to bypass the primary screening and grit removal facilities. An overflow outlet will be provided for the equalisation basin to prevent hydraulic overload during emergency situations.       |
| <b>Maintenance Accessibility</b>       | Screens, grit chambers, and the equalisation basin are designed with ease of access in mind, facilitating rapid maintenance and emergency interventions.                                                                                                 |
| <b>Emergency Procedures and Alarms</b> | An alarm system will trigger when blockage levels reach 50% or if there is a sudden halt in mechanical operation. This proactive alert mechanism enables prompt operator intervention, minimizing the risk of overflow or downstream process disruption. |

where  $E_r$  is the removal efficiency,  $T$  is the temperature of the water, and  $a, b$  are parameters defined for SST and BOD, see Table 9. The greater of the two detention times governs the total volume design.  $E_r$  was determined using Eq.(17), based on the inflow  $BOD$  and the effluent one that was found to comply with the environmental requirements, see Sections 4 and 5. The input values as well as the obtained time are shown in the Table 9. The retention time is checked against typical values, which are:  $1.5 \text{ h} < t < 2.5 \text{ h}$ , [Metcalf and Eddy, 2014].

$$E_r = \frac{BOD_{5,\text{influent}} - BOD_{5,\text{effluent}}}{BOD_{5,\text{influent}}} \times 100 \quad (17)$$

As there is no environmental requirement for the  $TSS$  content on the waterway discharge, the required efficiency can not be calculated. Instead, the efficiency and residence time are calculated to comply with the  $BOD_5$  content (Eq.(17)). The same detention time is assumed for the  $TSS$  and the efficiency for the solids removal is then calculated using the Eq.(17) with the corresponding parameters  $a$  and  $b$  from Table 9.

$$V_{\text{clarifier}}^{\text{calc}} = Q_{\text{peak}} \cdot t \cdot 3600 \quad (18)$$

Table 9: Standard parameters for detention time [Metcalf and Eddy, 2014].

| Constituent | a      | b     |
|-------------|--------|-------|
| $BOD_5$     | 0.018  | 0.020 |
| $TSS$       | 0.0075 | 0.014 |

The clarifier calculated volume ( $V_{\text{clarifier}}^{\text{calc}}$ ) is determined using Eq.(18) replacing the time  $t$  with  $t_{BOD}$  shown in Table 10. In addition, replacing the maximum and minimum typical retention times (see Ap-

pendix 2), the maximum and minimum primary sedimentation tank volume ( $V_{clarifier}^{min}$  and  $V_{clarifier}^{max}$ ) can be determine.

If  $V_{clarifier}^{calc}$  is greater than  $V_{clarifier}^{max}$ , then the excessive volume is used for the equalization basin ( $V_{equalization}^{req}$ ) and the final clarifier volume ( $V_{clarifier}^{final}$ ) will be equal to  $V_{clarifier}^{max}$ . Due to the equalization tank, there will be a new *adjusted* peak flow ( $Q_{peak}^{adj}$ ), obtained with Eq.(19). Based on this, subsequent calculations will be based on this value instead of the  $Q_{peak}$ , when required.

$$Q_{peak}^{adj} = \frac{V_{clarifier}^{max}}{t} \quad (19)$$

Table 10: Input values for clarifier sizing

| Parameter                | Value    | Units             | Description                                                           |
|--------------------------|----------|-------------------|-----------------------------------------------------------------------|
| $Q_{avg}$                | 4205.45  | m <sup>3</sup> /h | Average projected flow rate, see Table 3                              |
| $Q_{peak}$               | 7465.79  | m <sup>3</sup> /h | Peak projected flow rate, see Table 3                                 |
| $BOD_{5, influent}$      | 370.08   | mg/L              | Inflow $BOD_5$ , see Section 1.6                                      |
| $BOD_{5, effluent}$      | 259.26   | mg/L              | Found $BOD_5$ that complies with environmental impact, see Section 4  |
| $T$                      | 10       | C                 | Assumed water lowest Temperature to adjust retention time.            |
| $E_{BOD}$                | 30.0     | %                 | $BOD_5$ Efficiency removal, obtained with Eq.(17)                     |
| $t_{BOD}$                | 1.35     | h                 | Optained with Eq.(15)                                                 |
| $t_{TSS}$                | 1.35     | h                 | Same as $t_{BOD}$                                                     |
| $E_{TSS}$                | 51.14    | %                 | Solved from Eq.(17), replacing $t_{TSS}$                              |
| $V_{clarifiers}^{calc}$  | 13589.15 | m <sup>3</sup>    | Calculated with Eq.(18)                                               |
| $V_{clarifiers}^{max}$   | 10513.62 | m <sup>3</sup>    | Calculated with Eq.(18), using max. time residency.                   |
| $V_{equalization}^{req}$ | 3075.53  | m <sup>3</sup>    | Difference between $V_{clarifiers}^{max}$ and $V_{clarifiers}^{calc}$ |
| $Q_{peak}^{adj}$         | 5776.11  | m <sup>3</sup> /h | Adjusted peak flowrate, see Section 3.2                               |

The sedimentation basin height is then selected based on the typical design criteria, see Appendix 2. The 5 m side water depth height upper limit is selected and 0.5 m of freeboard is added ( $H_{clarifier} = 5.5m$ ). The area  $A_{clarifier}$  is calculated using Eq.(20), the volume  $V_{clarifier}^{final}$  used is equal to  $V_{clarifier}^{max}$  and the Surface Loading Rate ( $SLR$ ) with Eq.(21), using both the  $Q_{avg}$  and  $Q_{peak}$ . The clarifier diameter is then calculated using Eq.(22). The number of clarifiers  $n$  is selected equal as 3 to comply with the *Weir Loading Rate*, see next section. The clarifier volume  $V_{clarifier}^{final}$  will be supply in 3 equal volume tanks. Results are shown in Table 11, which are in compliance with the typical design criteria, see Appendix 2.

$$A_{clarifier} = \frac{V_{clarifier}^{final}}{n \cdot H} \quad (20)$$

$$SLR_{clarifier} = \frac{Q}{n \cdot A_{clarifier}} \cdot \frac{1}{3600 \cdot 24} \quad (21)$$

$$D_{clarifier} = \sqrt{\frac{4V_{clarifier}^{final}}{\pi \cdot n \cdot H_{clarifier}}} \quad (22)$$

### 3.2.2 Weir Loading Rate (WLR) Assessment

The weir length for circular clarifiers with two-sided launders is given by Eq.(23), where  $d_{offset}$  is the distance from clarifier edge to centerline of the launder and  $n_{launder-sides}$  is the number of launder sides. The Weir Length Ratio ( $WLR$ ) is given by Ec.(24). The critical condition is then evaluated with Eq.(25), where  $n$  is the number of clarifier tanks. As explained in the previous section, this value was set as 3 so  $WLR_{critical}$  does not exceed the design limit (500 m<sup>3</sup>/m/day).

$$L_w = 2 \cdot \pi \cdot (D_{clarifier} - n_{launder-sides} \cdot d_{offset}) \quad (23)$$

$$WLR = \frac{Q_{peak}}{L_w \cdot 86400} \quad (24)$$

$$WLR_{critical} = \frac{WLR}{n - 1} \quad (25)$$

### 3.2.3 Final Configuration and Results

The process abovementioned was iterated until achieving results that comply with the typical design criteria (Appendix 2), result are shown in the Table 11.

Table 11: Final clarifier design parameters

| Parameter                | Result  | Units                               | Description                                   |
|--------------------------|---------|-------------------------------------|-----------------------------------------------|
| $A_{clarifier}$          | 700.91  | m <sup>2</sup>                      | Area per clarifier.                           |
| $SLR_{clarifier}^{avg}$  | 48.0    | m <sup>3</sup> /m <sup>2</sup> /day | Surface Loading Rate under average flow rate. |
| $SLR_{clarifier}^{peak}$ | 65.93   | m <sup>3</sup> /m <sup>2</sup> /day | Surface Loading Rate under peak flow rate.    |
| $D_{clarifier}$          | 29.87   | m                                   | Diameter of clarifier tank.                   |
| $n$                      | 3       | –                                   | Number of clarifier tanks.                    |
| $n_{launder-sides}$      | 2       | –                                   | Number of launder sides.                      |
| $L_w$                    | 175.13  | m                                   | Weir length.                                  |
| $WLR$                    | 791.55  | m <sup>3</sup> /m/day               | Weir Lenth Ratio.                             |
| $WLR_{critical}$         | 395.77  | m <sup>3</sup> /m/day               | Weir Lenth Ratio under critial condition.     |
| $V_{equalization}^{req}$ | 3075.53 | m <sup>3</sup>                      | Required equilization tank volume.            |

### 3.3 V-Notch and Launder Design

To ensure stable discharge from the primary clarifiers and prevent hydraulic overloading, a V-notch weir system was designed in accordance with standard hydraulic principles. The number of notches is determined based on the critical launder length and a center-to-center spacing of 0.3 m between notches:

$$N_{\text{notch}} = \left\lceil \frac{L_{\text{launder}}}{s} \right\rceil \quad (26)$$

where  $s = 0.3$  m is the spacing. The total number of V-notches in the system is then:

$$N_{\text{notch, total}} = N_{\text{notch}} \times N_{\text{clarifiers}} \quad (27)$$

The flow per notch is calculated based on the design peak hourly flow:

$$Q_{\text{notch}} = \frac{Q_{\text{peak}}^{\text{adj}}}{N_{\text{notch, total}}} \quad (28)$$

The maximum height of water over each V-notch is determined using the general formula for triangular weirs:

$$h = \left( \frac{Q_{\text{notch}}}{\frac{8}{15} C_d \sqrt{2g} \tan(\theta/2)} \right)^{2/5} \quad (29)$$

where  $C_d = 0.58$ ,  $g = 9.81$  m/s<sup>2</sup>, and  $\theta = 90^\circ$ . A 15% safety margin is applied to determine the actual design depth above the notch:

$$y = 1.15 \cdot h \quad (30)$$

The top width of the V-notch opening is twice the water depth:

$$w_{\text{notch}} = 2y \quad (31)$$

### 3.3.1 Launder Flow Conditions

To assess flow behavior in the launder channel, the discharge per unit length of the launder is:

$$q = W L R_{\text{critical}} \cdot \frac{1}{3600 \cdot 24} \cdot n_{\text{launder-sides}} \quad (32)$$

The maximum water depth in the launder is calculated from the critical section at the discharge point:

$$Y_c = \left( \frac{q^2 \cdot L_{\text{launder}}^2}{4g w_{\text{notch}}^2} \right)^{1/3} \quad (33)$$

The perimeter of a clarifier with a 2-sided launder can be determined by dividing the total weir length ( $L_w$ ) by the number of launder sides:

$$P_{\text{clarifier}} = \frac{L_w}{n_{\text{laundry-sides}}} \quad (34)$$

The farthest distance that water travels into the launder, denoted as  $x$ , is half the perimeter divided by the number of discharge points:

$$x = \frac{P_{\text{clarifier}}}{2 \cdot n_{\text{disch}}} \quad (35)$$

The final maximum water depth is obtained using the equation for uniform flow depth in a launder with one outlet point ( $n_{\text{disch}} = 1$ ) at distance  $x$ :

$$H = \sqrt{Y_c^2 + \frac{2q^2x^2}{gw^2Y_c}} \quad (36)$$

A 50% safety factor is applied to this maximum depth for design purposes:

$$H_{\text{final}} = 1.5 \cdot H \quad (37)$$

### 3.3.2 V-Notch and Launder design results

In the Table 12 the results of V-Notch and Launder design are presented.

*Table 12: V-notch and launder design summary*

| Parameter                    | Result  | Units               | Description                                             |
|------------------------------|---------|---------------------|---------------------------------------------------------|
| $N_{\text{notch/clarifier}}$ | 584.0   | –                   | Number of notches per clarifier.                        |
| $N_{\text{notch, total}}$    | 1752.0  | –                   | Total number of notches across all clarifiers.          |
| $Q_{\text{notch}}$           | 0.00092 | m <sup>3</sup> /s   | Flow per individual V-notch.                            |
| $h$                          | 0.0537  | m                   | Maximum height of water over the notch.                 |
| $y$                          | 0.06176 | m                   | Water depth over the notch including 15% safety factor. |
| $w_{\text{notch}}$           | 0.124   | m                   | Top width of the V-notch.                               |
| $q$                          | 0.00916 | m <sup>3</sup> /m/s | Discharge per unit length of launder.                   |
| $Y_c$                        | 0.403   | m                   | Critical height at launder discharge point.             |
| $H$                          | 0.699   | m                   | Maximum calculated water depth in launder.              |
| $H_{\text{final}}$           | 1.048   | m                   | Final design depth in launder with 50% safety factor.   |

### 3.4 Protection and future considerations

To ensure reliability and continuity of treatment, the primary clarifier system is designed with redundancy and fail-safes. Specifically, multiple clarifier units are provided such that one can be taken offline for maintenance without disrupting treatment capacity, that is why on Section 3.2  $n - 1$  was used for the weir design. Inlet flow is evenly distributed via energy-dissipating inlets to prevent localized overloading. Overflow weirs are designed with excess capacity and freeboard to safely manage flow surges and prevent bypassing. Emergency bypass channels are also integrated into the layout to protect downstream processes during extreme hydraulic loads or clarifier failure events.

## 4 SECONDARY TREATMENT

### 4.1 Design Overview

The secondary treatment stage is designed as an extended aeration activated sludge process, aiming to achieve a 94.6 %  $BOD_5$  removal efficiency. As shown in Section 5, in order to comply with the minimum  $DO$  in the waterway, the maximum  $BOD$  present in the effluent must be equal to 14 mg/L. Based on that and the removal efficiency, the inflow  $BOD$  concentration can be calculated as:

$$BOD_{in} = \frac{BOD_{effluent}}{1 - \eta} \quad (38)$$

where  $\eta$  is the BOD removal efficiency. Computed, the  $BOD_{in}$  is equal to 259.3 mg/L, and it represents the target for the removal in the primary treatment, see Section 3.

### 4.2 Sludge Characteristics and Return Flow

Mixed liquor suspended solids (MLSS) concentration is chosen based on the process type and operating temperature, and return sludge concentration  $X_R$  is assumed. The sludge volume index (SVI) is given by:

$$SVI = \frac{10^6}{X_R} \quad (39)$$

The settled sludge volume is:

$$v = \frac{SVI \cdot X}{1000} \quad (40)$$

The return sludge flow rate is then:

$$Q_R = \frac{Q \cdot v}{1000 - v} \quad (41)$$

## 4.3 Aeration Tank Sizing

Using a target food-to-microorganism ratio ( $F/M$ ), the required aeration tank volume is calculated:

$$V_{\text{aeration}} = \frac{Q \cdot BOD_{\text{in}}}{F/M \cdot X} \quad (42)$$

This is followed by residence time:

$$t_{\text{aeration}} = \frac{V_{\text{aeration}}}{Q} \cdot 24 \quad (43)$$

The organic loading is:

$$OL = \frac{Q \cdot BOD_{\text{in}}}{V_{\text{aeration}}} \quad (44)$$

## 4.4 Sludge Age and Stability

The sludge age is estimated from:

$$\theta_c = \frac{1}{Y \cdot (F/M) \cdot \eta - K_d} \quad (45)$$

A safety factor is applied to account for peak organic load variations:

$$\theta_{c,\text{design}} = 1.3 \cdot \theta_c \quad (46)$$

## 4.5 Contact-Stabilization Configuration

Given the contact time is 0.5 h and stabilization time is 2 h, the aeration tank is split as follows:

$$V_{\text{contact}} = \frac{1}{5} \cdot V_{\text{aeration}} \quad (47)$$

$$V_{\text{stabilization}} = V_{\text{aeration}} - V_{\text{contact}} \quad (48)$$

Contact tank geometry is based on the guideline ratio length =  $4 \times$  depth, and width computed from:

$$W = \frac{V}{L \cdot D} \quad (49)$$

Stabilization tank depth is derived from:

$$D = \frac{V}{L \cdot W} \quad (50)$$

Table 13: Input values for secondary treatment

| Parameter                        | Symbol                  | Value                |
|----------------------------------|-------------------------|----------------------|
| Target effluent BOD <sub>5</sub> | $BOD_{\text{effluent}}$ | 20 mg/L              |
| BOD removal efficiency           | $\eta$                  | 90%                  |
| MLSS                             | $X$                     | 3000 mg/L            |
| Return sludge solids             | $X_R$                   | 11000 mg/L           |
| Yield coefficient                | $Y$                     | 0.6                  |
| Decay coefficient                | $K_d$                   | 0.06 d <sup>-1</sup> |
| Conversion rate                  | $k$                     | 0.2 d <sup>-1</sup>  |
| F/M ratio                        | F/M                     | 0.4                  |
| Safety factor                    | –                       | 1.3                  |

Table 14: Design results for secondary treatment

| Output                     | Symbol                     | Result                               |
|----------------------------|----------------------------|--------------------------------------|
| Influent BOD to secondary  | $BOD_{\text{in}}$          | [insert]                             |
| Return sludge ratio        | $Q_R/Q$                    | [insert]                             |
| Aeration volume            | $V_{\text{aeration}}$      | [insert] m <sup>3</sup>              |
| Aeration time              | $t_{\text{aeration}}$      | [insert] h                           |
| Sludge age                 | $\theta_c$                 | [insert] d                           |
| Final design age           | $\theta_{c,\text{design}}$ | [insert] d                           |
| Contact tank volume        | $V_{\text{contact}}$       | [insert] m <sup>3</sup>              |
| Stabilization tank volume  | $V_{\text{stabilization}}$ | [insert] m <sup>3</sup>              |
| Contact tank (L×W×D)       | –                          | [insert] m × [insert] m × [insert] m |
| Stabilization tank (L×W×D) | –                          | [insert] m × [insert] m × [insert] m |

## Summary of Inputs and Outputs

## 5 ENVIRONMENTAL IMPACT ASSESSMENT

To evaluate the impact of the effluent discharge on the receiving waterway, the *Streeter-Phelps model* was applied. This model predicts the variation in dissolved oxygen (DO) downstream of the discharge point based on oxygen consumption due to biochemical oxygen demand (BOD) and oxygen replenishment through atmospheric reaeration.

According to previous surveys and analysis, the receiving waterway has a base flow that complies with:  $Q_{\text{waterway}} = 300,000 \cdot (10)^{0.2} = 475468 \text{ m}^3/\text{day}$ . The water quality upstream of the discharge point is characterized by a background BOD of 4 mg/L ( $L_r$ ), a dissolved oxygen saturation of 8.5 mg/L ( $DO_{\text{saturation}}$ ), and an initial oxygen deficit of 0.8 mg/L ( $DO_0$ ). Regulatory guidelines for this region require that DO levels remain above 70% saturation **REFERENCE TBD**, yielding a minimum acceptable DO of 5.95 mg/L (Eq (51)). Additionally, the reaeration rate constant ( $k_r$ ), deoxygenation rate constant ( $k_d$ ), BOD reaction rate ( $k_1$ ) and stream velocity ( $v$ ) are known due to previous assessments. In the Table 15 the input values for the *Streeter-Phelps model* are shown.

The effluent discharge was evaluated based on the project adjusted peak flow rate ( 138626.63 m<sup>3</sup>/day), calculated in Section 3 and a effluent BOD<sub>5</sub> of 14 mg/L ( $L_w$ ). This value was estimated through iterating the calculations explained in this sections until achieving a  $DO_{\text{calculated}}$  that complies with the

$DO_{min, threshold}$ . Thus, this value was set as the target for the upstream processes, see Sections 3 and 4.

Table 15: Input values for Streeter-Phelps DO sag model

| Parameter             | Value                         | Description                  |
|-----------------------|-------------------------------|------------------------------|
| $DO_{saturation}$     | 8.5 mg/L                      | DO at saturation             |
| $DO_0$                | 0.8 mg/L                      | Initial oxygen deficit       |
| $DO_{min, threshold}$ | 5.95 mg/L                     | Minimum DO required          |
| $L_r$                 | 4 mg/L                        | BOD upstream                 |
| $L_w$                 | 14 mg/L                       | BOD <sub>5</sub> of effluent |
| $k_r$                 | 0.71 day <sup>-1</sup>        | Reaeration rate constant     |
| $k_d$                 | 0.51 day <sup>-1</sup>        | Deoxygenation rate constant  |
| $k_1$                 | 0.23 day <sup>-1</sup>        | BOD reaction rate            |
| $v$                   | 0.07 m/s                      | Stream velocity              |
| $Q_w$                 | 475468 m <sup>3</sup> /day    | Waterway flowrate            |
| $Q_e$                 | 138626.63 m <sup>3</sup> /day | Adjusted flowrate            |

From this, the ultimate BOD ( $L_u$ ) is estimated using Eq. (52) with the time  $t$  equals to 5 days. Using Eq. (53) the mixed BOD in the receiving stream is determined, and with Eq. (53) mixed conditions downstream are calculated. The model then determines the time ( $t_c$ ) at which the minimum DO level occurs using Eq. (54). Finally, the maximum deficit  $D(t_c)$  is calculated using Eq. (56). The distance downstream at which the the maximum deficit/minimum saturation is reached can also be calculated using Eq. (55) Results are shown in Table 16.

$$DO_{min, threshold} = 0.7 \cdot DO_{saturation} \quad (51)$$

$$L_u = \frac{L_w}{1 - e^{-k_1 t}} \quad (52)$$

$$L_0 = \frac{Q_w \cdot L_r + Q_e \cdot L_u}{Q_w + Q_e} \quad (53)$$

$$t_c = \frac{1}{k_r - k_d} \ln \left[ \frac{k_r}{k_d} \left( 1 - D_0 \cdot \frac{k_r - k_d}{k_d L_0} \right) \right] \quad (54)$$

$$x_c = t_c \cdot 86400 \cdot v \quad (55)$$

$$D(t_c) = \frac{k_d L_0}{k_r - k_d} \left( e^{-k_d t_c} - e^{-k_r t_c} \right) + D_0 e^{-k_r t_c} \quad (56)$$

$$DO_{calculated} = DO_{saturation} - D(t_c) \quad (57)$$

As shown in Table 16, the dissolved oxygen deficit at its critical point is 2.5 mg/L. Subtracting from the saturation  $DO_{saturation}$  the critical deficit  $D(t_c)$  the  $DO_{calculated}$  is obtained, equal to 6.0 mg/L which is above the regulatory  $DO_{min, threshold}$  of 5.95 mg/L. Therefore, the discharge complies with the receiving waterway's environmental standard.

In the Figure 5 the deficit variation across time can be observed, obtained base on Eq. (56) varying  $t$ . The vertical axes represents the deficit, thus zero value indicates the waterway returning to a  $DO_{saturation}$  level. In the figure, it can be observed the maximum deficit which is achieved before  $t = 2days$ . It can also be observed that after approximately 5 days the  $DO$  returns to the initial waterway

Table 16: Environmental impact model results

| Output            | Value      | Description                                  |
|-------------------|------------|----------------------------------------------|
| $L_u$             | 20.49 mg/L | Ultimate BOD of effluent                     |
| $L_0$             | 7.24 mg/L  | Mixed BOD in receiving stream                |
| $t_c$             | 1.43 days  | Time to reach minimum DO                     |
| $x_c$             | 8665.43 m  | Distance to minimum DO level                 |
| $D(t_c)$          | 2.5 mg/L   | DO deficit at time $t_c$                     |
| $DO_{calculated}$ | 6.0 mg/L   | DO in waterway when maximum deficit $D(t_c)$ |

upstream initial value ( $DO_0$ ).

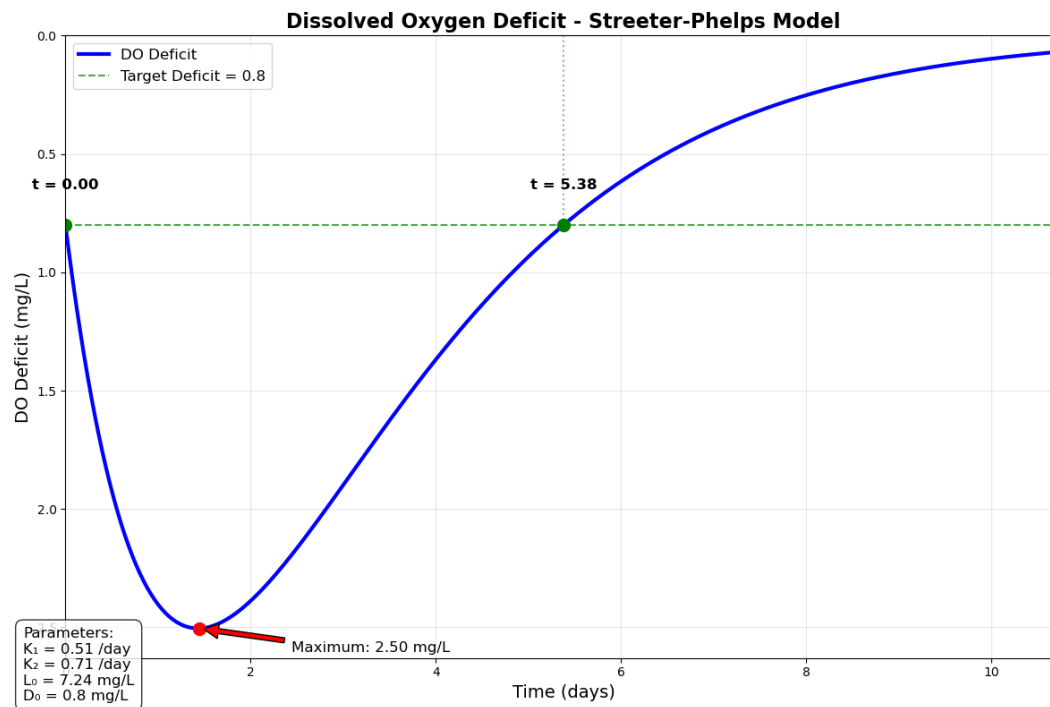


Figure 5: Dissolved oxygen deficit. Vertical axes being the deficit, with zero representation saturation.

## **6 SUMMARY**

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## APPENDIX 1 AREA AND POPULATION PER SUBURB

Table of Melbourne suburbs with their area and population [Australian Bureau of Statistics, 2021].

*Table 17: Melbourne suburbs with area and population*

| Suburb          | Population Density | Population | LGA                  | Area (km2) |
|-----------------|--------------------|------------|----------------------|------------|
| Southbank       | 11962.280000       | 22631      | Melbourne (City)     | 1.890000   |
| Carlton         | 10597.480000       | 16055      | Melbourne (City)     | 1.510000   |
| Fitzroy         | 7355.630000        | 10431      | Yarra (City)         | 1.420000   |
| Melbourne       | 7269.020000        | 16055      | Melbourne (City)     | 2.210000   |
| South Yarra     | 7087.660000        | 25028      | Melbourne (City)     | 3.530000   |
| Balaclava       | 7062.830000        | 5392       | Port Phillip (City)  | 0.760000   |
| Prahran         | 6975.820000        | 12203      | Stonnington (City)   | 1.750000   |
| Windsor         | 6729.210000        | 7273       | Port Phillip (City)  | 1.080000   |
| Collingwood     | 6719.020000        | 9179       | Yarra (City)         | 1.370000   |
| St Kilda        | 6365.640000        | 19490      | Port Phillip (City)  | 3.060000   |
| North Melbourne | 6325.150000        | 14953      | Melbourne (City)     | 2.360000   |
| Richmond        | 6252.540000        | 28587      | Yarra (City)         | 4.570000   |
| Elwood          | 5989.600000        | 15153      | Port Phillip (City)  | 2.530000   |
| St Kilda West   | 5977.320000        | 2951       | Port Phillip (City)  | 0.490000   |
| Travancore      | 5933.010000        | 2116       | Moonee Valley (City) | 0.360000   |
| St Kilda East   | 5784.110000        | 12571      | Glen Eira (City)     | 2.170000   |
| Glen Huntly     | 5637.580000        | 4905       | Glen Eira (City)     | 0.870000   |
| Seddon          | 5586.700000        | 5143       | Maribyrnong (City)   | 0.920000   |
| Ripponlea       | 5510.490000        | 1532       | Port Phillip (City)  | 0.280000   |
| Kingsville      | 5442.760000        | 3920       | Maribyrnong (City)   | 0.720000   |
| Brunswick East  | 5097.030000        | 13279      | Moreland (City)      | 2.610000   |
| Kensington      | 5073.670000        | 10745      | Melbourne (City)     | 2.120000   |
| Fitzroy North   | 5007.710000        | 12781      | Moreland (City)      | 2.550000   |
| South Melbourne | 4986.300000        | 11548      | Port Phillip (City)  | 2.320000   |
| Brunswick       | 4910.310000        | 24896      | Moreland (City)      | 5.070000   |
| Princes Hill    | 4909.930000        | 2005       | Yarra (City)         | 0.410000   |
| Middle Park     | 4885.610000        | 4000       | Port Phillip (City)  | 0.820000   |
| Carnegie        | 4716.030000        | 17909      | Glen Eira (City)     | 3.800000   |
| Abbotsford      | 4700.750000        | 9088       | Yarra (City)         | 1.930000   |
| Brunswick West  | 4357.960000        | 14746      | Moreland (City)      | 3.380000   |
| Armada          | 4201.390000        | 9368       | Stonnington (City)   | 2.230000   |
| Hawthorn East   | 4110.510000        | 14834      | Boroondara (City)    | 3.610000   |
| Northcote       | 4087.370000        | 25276      | Darebin (City)       | 6.180000   |
| Essendon North  | 4080.600000        | 3071       | Moonee Valley (City) | 0.750000   |
| Ormond          | 4064.220000        | 8328       | Glen Eira (City)     | 2.050000   |
| Hawthorn        | 4000.510000        | 22322      | Boroondara (City)    | 5.580000   |
| Elsternwick     | 3975.800000        | 10887      | Glen Eira (City)     | 2.740000   |

Continued on next page

Table 17: Melbourne suburbs with area and population

| Suburb            | Population Density | Population | LGA                      | Area (km2) |
|-------------------|--------------------|------------|--------------------------|------------|
| McKinnon          | 3897.170000        | 6878       | Glen Eira (City)         | 1.760000   |
| Gardenvale        | 3884.170000        | 1019       | Glen Eira (City)         | 0.260000   |
| Ascot Vale        | 3845.150000        | 15197      | Moonee Valley (City)     | 3.950000   |
| Caulfield         | 3806.120000        | 5748       | Glen Eira (City)         | 1.510000   |
| Coburg            | 3785.050000        | 26574      | Moreland (City)          | 7.020000   |
| Murrumbeena       | 3771.280000        | 9996       | Glen Eira (City)         | 2.650000   |
| Hughesdale        | 3759.200000        | 7563       | Monash (City)            | 2.010000   |
| Clifton Hill      | 3646.350000        | 6606       | Yarra (City)             | 1.810000   |
| Caulfield North   | 3633.750000        | 16903      | Glen Eira (City)         | 4.650000   |
| Noble Park        | 3624.230000        | 32257      | Greater Dandenong (City) | 8.900000   |
| Caulfield South   | 3621.750000        | 12328      | Glen Eira (City)         | 3.400000   |
| Thornbury         | 3617.380000        | 19005      | Darebin (City)           | 5.250000   |
| Docklands         | 3482.850000        | 15495      | Melbourne (City)         | 4.450000   |
| Malvern           | 3453.170000        | 9929       | Stonnington (City)       | 2.880000   |
| Bentleigh         | 3426.600000        | 17921      | Glen Eira (City)         | 5.230000   |
| Pascoe Vale       | 3423.210000        | 18171      | Moreland (City)          | 5.310000   |
| Carlton North     | 3367.180000        | 6177       | Melbourne (City)         | 1.830000   |
| Essendon          | 3344.050000        | 21240      | Moonee Valley (City)     | 6.350000   |
| Pascoe Vale South | 3309.990000        | 10534      | Moreland (City)          | 3.180000   |
| Hampton East      | 3295.150000        | 5069       | Bayside (City)           | 1.540000   |
| Parkdale          | 3292.040000        | 12308      | Kingston (City)          | 3.740000   |
| Moonee Ponds      | 3272.100000        | 16224      | Moonee Valley (City)     | 4.960000   |
| Taylors Hill      | 3254.310000        | 15419      | Melton (City)            | 4.740000   |
| Kings Park        | 3242.880000        | 8203       | Brimbank (City)          | 2.530000   |
| Footscray         | 3236.630000        | 17131      | Maribyrnong (City)       | 5.290000   |
| Box Hill          | 3235.380000        | 14353      | Whitehorse (City)        | 4.440000   |
| Meadow Heights    | 3209.080000        | 14890      | Hume (City)              | 4.640000   |
| Roxburgh Park     | 3171.080000        | 24129      | Hume (City)              | 7.610000   |
| Seabrook          | 3162.960000        | 4952       | Hobsons Bay (City)       | 1.570000   |
| Oakleigh East     | 3131.200000        | 6804       | Monash (City)            | 2.170000   |
| Hampton           | 3120.720000        | 13518      | Bayside (City)           | 4.330000   |
| Niddrie           | 3083.290000        | 5901       | Moonee Valley (City)     | 1.910000   |
| Glen Iris         | 3069.860000        | 26131      | Boroondara (City)        | 8.510000   |
| Surrey Hills      | 3069.720000        | 13655      | Boroondara (City)        | 4.450000   |
| Bentleigh East    | 3057.980000        | 30159      | Glen Eira (City)         | 9.860000   |
| West Footscray    | 3056.590000        | 11729      | Maribyrnong (City)       | 3.840000   |
| Oak Park          | 3053.640000        | 6714       | Moreland (City)          | 2.200000   |
| Sydenham          | 3044.380000        | 10578      | Brimbank (City)          | 3.470000   |
| Burnside Heights  | 3043.610000        | 6377       | Melton (City)            | 2.100000   |
| Balwyn            | 3026.140000        | 13495      | Boroondara (City)        | 4.460000   |
| Toorak            | 3021.060000        | 12817      | Stonnington (City)       | 4.240000   |
| Blackburn South   | 3000.560000        | 10939      | Whitehorse (City)        | 3.650000   |

Continued on next page

Table 17: Melbourne suburbs with area and population

| Suburb             | Population Density | Population | LGA                      | Area (km <sup>2</sup> ) |
|--------------------|--------------------|------------|--------------------------|-------------------------|
| Box Hill North     | 2993.190000        | 12337      | Whitehorse (City)        | 4.120000                |
| Cremorne           | 2967.650000        | 2158       | Yarra (City)             | 0.730000                |
| Kingsbury          | 2964.710000        | 3460       | Darebin (City)           | 1.170000                |
| Chelsea            | 2961.140000        | 8347       | Kingston (City)          | 2.820000                |
| Albanvale          | 2947.400000        | 5641       | Brimbank (City)          | 1.910000                |
| Lalor              | 2924.790000        | 23219      | Whittlesea (City)        | 7.940000                |
| Heidelberg Heights | 2916.630000        | 6758       | Banyule (City)           | 2.320000                |
| Camberwell         | 2909.610000        | 21965      | Boroondara (City)        | 7.550000                |
| Caroline Springs   | 2904.020000        | 24488      | Melton (City)            | 8.430000                |
| St Albans          | 2899.360000        | 38042      | Brimbank (City)          | 13.120000               |
| Preston            | 2864.580000        | 33790      | Darebin (City)           | 11.800000               |
| Mentone            | 2863.300000        | 13197      | Kingston (City)          | 4.610000                |
| Maidstone          | 2858.230000        | 9389       | Maribyrnong (City)       | 3.280000                |
| Highett            | 2857.850000        | 12016      | Bayside (City)           | 4.200000                |
| Mont Albert        | 2843.710000        | 4948       | Boroondara (City)        | 1.740000                |
| Brighton East      | 2823.010000        | 16757      | Bayside (City)           | 5.940000                |
| Edithvale          | 2811.620000        | 6276       | Kingston (City)          | 2.230000                |
| Malvern East       | 2811.060000        | 22296      | Stonnington (City)       | 7.930000                |
| Dallas             | 2808.250000        | 6762       | Hume (City)              | 2.410000                |
| Springvale South   | 2782.910000        | 12766      | Greater Dandenong (City) | 4.590000                |
| Brighton           | 2778.140000        | 23252      | Bayside (City)           | 8.370000                |
| Chadstone          | 2769.550000        | 9552       | Monash (City)            | 3.450000                |
| Sandringham        | 2769.330000        | 10926      | Bayside (City)           | 3.950000                |
| Blackburn North    | 2744.920000        | 7627       | Whitehorse (City)        | 2.780000                |
| Ashwood            | 2705.700000        | 7154       | Monash (City)            | 2.640000                |
| Flemington         | 2687.670000        | 15197      | Melbourne (City)         | 5.650000                |
| Fawkner            | 2685.090000        | 14274      | Hume (City)              | 5.320000                |
| East Melbourne     | 2674.570000        | 4896       | Melbourne (City)         | 1.830000                |
| South Kingsville   | 2663.470000        | 2156       | Hobsons Bay (City)       | 0.810000                |
| Reservoir (Vic.)   | 2661.150000        | 51096      | Darebin (City)           | 19.200000               |
| Ashburton          | 2651.730000        | 7952       | Boroondara (City)        | 3.000000                |
| Gowanbrae          | 2645.990000        | 2971       | Moreland (City)          | 1.120000                |
| Yarraville         | 2644.460000        | 15636      | Maribyrnong (City)       | 5.910000                |
| Dandenong          | 2636.980000        | 30127      | Greater Dandenong (City) | 11.420000               |
| Watsonia North     | 2630.340000        | 3799       | Banyule (City)           | 1.440000                |
| Forest Hill        | 2610.810000        | 10780      | Whitehorse (City)        | 4.130000                |
| Beaumaris          | 2588.020000        | 13947      | Bayside (City)           | 5.390000                |
| Newport            | 2580.100000        | 13658      | Hobsons Bay (City)       | 5.290000                |
| Canterbury         | 2573.800000        | 7800       | Boroondara (City)        | 3.030000                |
| Carrum             | 2559.490000        | 4239       | Kingston (City)          | 1.660000                |
| Doncaster East     | 2553.110000        | 30926      | Manningham (City)        | 12.110000               |
| Mont Albert North  | 2542.980000        | 5609       | Whitehorse (City)        | 2.210000                |

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Table 17: Melbourne suburbs with area and population

| Suburb             | Population Density | Population | LGA                      | Area (km2) |
|--------------------|--------------------|------------|--------------------------|------------|
| Rosanna            | 2518.670000        | 8616       | Banyule (City)           | 3.420000   |
| Aberfeldie         | 2510.950000        | 3925       | Moonee Valley (City)     | 1.560000   |
| Williamstown       | 2504.300000        | 14407      | Hobsons Bay (City)       | 5.750000   |
| Clayton            | 2475.130000        | 18988      | Monash (City)            | 7.670000   |
| Burwood            | 2469.010000        | 15147      | Monash (City)            | 6.130000   |
| Strathmore         | 2438.170000        | 8980       | Moonee Valley (City)     | 3.680000   |
| Mitcham            | 2436.700000        | 16795      | Whitehorse (City)        | 6.890000   |
| Box Hill South     | 2416.620000        | 8491       | Whitehorse (City)        | 3.510000   |
| Glenroy            | 2411.640000        | 23792      | Moreland (City)          | 9.870000   |
| Burwood East       | 2409.810000        | 10675      | Whitehorse (City)        | 4.430000   |
| Glen Waverley      | 2394.710000        | 42642      | Monash (City)            | 17.810000  |
| Delahey            | 2378.490000        | 8077       | Brimbank (City)          | 3.400000   |
| Dandenong North    | 2371.500000        | 22550      | Greater Dandenong (City) | 9.510000   |
| Doncaster          | 2366.780000        | 11219      | Manningham (City)        | 4.740000   |
| Blackburn          | 2363.110000        | 14478      | Whitehorse (City)        | 6.130000   |
| Briar Hill         | 2354.000000        | 3220       | Banyule (City)           | 1.370000   |
| Kew                | 2346.690000        | 24499      | Boroondara (City)        | 10.440000  |
| Nunawading         | 2338.260000        | 12413      | Manningham (City)        | 5.310000   |
| Montmorency        | 2338.200000        | 9250       | Banyule (City)           | 3.960000   |
| Huntingdale        | 2333.330000        | 1949       | Monash (City)            | 0.840000   |
| Deepdene           | 2325.710000        | 2101       | Boroondara (City)        | 0.900000   |
| Doveton            | 2297.010000        | 9603       | Casey (City)             | 4.180000   |
| Bonbeach           | 2293.070000        | 6855       | Kingston (City)          | 2.990000   |
| Aspendale          | 2291.190000        | 7285       | Kingston (City)          | 3.180000   |
| Narre Warren South | 2290.990000        | 30909      | Casey (City)             | 13.490000  |
| Templestowe Lower  | 2285.400000        | 14098      | Manningham (City)        | 6.170000   |
| Mill Park          | 2273.840000        | 28712      | Whittlesea (City)        | 12.630000  |
| Oakleigh           | 2270.060000        | 8442       | Monash (City)            | 3.720000   |
| Watsonia           | 2265.970000        | 5352       | Banyule (City)           | 2.360000   |
| Vermont            | 2265.080000        | 10993      | Maroondah (City)         | 4.850000   |
| Balwyn North       | 2244.140000        | 21302      | Boroondara (City)        | 9.490000   |
| Maribyrnong        | 2223.920000        | 12573      | Maribyrnong (City)       | 5.650000   |
| Mount Waverley     | 2213.430000        | 35340      | Monash (City)            | 15.970000  |
| Ivanhoe            | 2196.930000        | 13374      | Banyule (City)           | 6.090000   |
| Ringwood East      | 2194.780000        | 10764      | Maroondah (City)         | 4.900000   |
| Cairnlea           | 2191.780000        | 10038      | Brimbank (City)          | 4.580000   |
| Albert Park        | 2189.150000        | 6044       | Port Phillip (City)      | 2.760000   |
| Heathmont          | 2171.220000        | 9933       | Maroondah (City)         | 4.570000   |
| Hoppers Crossing   | 2159.050000        | 37216      | Wyndham (City)           | 17.240000  |
| Gladstone Park     | 2157.870000        | 8213       | Hume (City)              | 3.810000   |
| Avondale Heights   | 2150.670000        | 12388      | Moonee Valley (City)     | 5.760000   |
| Macleod            | 2137.170000        | 9892       | Banyule (City)           | 4.630000   |

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Table 17: Melbourne suburbs with area and population

| Suburb            | Population Density | Population | LGA                      | Area (km2) |
|-------------------|--------------------|------------|--------------------------|------------|
| Croydon Hills     | 2136.070000        | 4839       | Maroondah (City)         | 2.270000   |
| Clarinda          | 2131.950000        | 7441       | Kingston (City)          | 3.490000   |
| Braybrook         | 2126.010000        | 9682       | Maribyrnong (City)       | 4.550000   |
| Heidelberg        | 2121.680000        | 7360       | Banyule (City)           | 3.470000   |
| Deer Park         | 2113.570000        | 18145      | Brimbank (City)          | 8.590000   |
| Cheltenham        | 2107.700000        | 13947      | Bayside (City)           | 6.620000   |
| Lynbrook          | 2093.120000        | 9121       | Casey (City)             | 4.360000   |
| Hillside          | 2078.730000        | 17331      | Brimbank (City)          | 8.340000   |
| Taylors Lakes     | 2075.840000        | 15174      | Brimbank (City)          | 7.310000   |
| Sunshine          | 2064.240000        | 9445       | Brimbank (City)          | 4.580000   |
| Airport West      | 2058.230000        | 8173       | Moonee Valley (City)     | 3.970000   |
| Keilor Lodge      | 2037.210000        | 1668       | Brimbank (City)          | 0.820000   |
| Seaholme          | 2035.900000        | 2067       | Hobsons Bay (City)       | 1.020000   |
| Greensborough     | 2030.520000        | 21070      | Banyule (City)           | 10.380000  |
| Aspendale Gardens | 2015.430000        | 6427       | Kingston (City)          | 3.190000   |
| Sunshine West     | 1995.490000        | 18552      | Brimbank (City)          | 9.300000   |
| Keilor Downs      | 1992.620000        | 9857       | Brimbank (City)          | 4.950000   |
| Noble Park North  | 1991.470000        | 7436       | Greater Dandenong (City) | 3.730000   |
| Cranbourne North  | 1964.060000        | 24683      | Casey (City)             | 12.570000  |
| Croydon North     | 1961.570000        | 8092       | Maroondah (City)         | 4.130000   |
| Springvale        | 1958.690000        | 22174      | Greater Dandenong (City) | 11.320000  |
| Boronia           | 1957.230000        | 23607      | Knox (City)              | 12.060000  |
| Jacana            | 1934.550000        | 2187       | Hume (City)              | 1.130000   |
| Black Rock        | 1933.750000        | 6389       | Bayside (City)           | 3.300000   |
| Bellfield         | 1925.890000        | 1996       | Banyule (City)           | 1.040000   |
| Notting Hill      | 1918.240000        | 2895       | Monash (City)            | 1.510000   |
| Hampton Park      | 1913.510000        | 26082      | Casey (City)             | 13.630000  |
| Croydon           | 1911.060000        | 28608      | Maroondah (City)         | 14.970000  |
| Albion            | 1905.360000        | 4334       | Brimbank (City)          | 2.270000   |
| Wheelers Hill     | 1892.230000        | 20652      | Monash (City)            | 10.910000  |
| Vermont South     | 1888.420000        | 11954      | Whitehorse (City)        | 6.330000   |
| Ferntree Gully    | 1887.580000        | 27398      | Knox (City)              | 14.510000  |
| Ringwood North    | 1880.280000        | 9964       | Manningham (City)        | 5.300000   |
| Burnside          | 1861.680000        | 5800       | Melton (City)            | 3.120000   |
| Fairfield         | 1853.590000        | 6535       | Darebin (City)           | 3.530000   |
| Mordialloc        | 1852.960000        | 8886       | Kingston (City)          | 4.800000   |
| Narre Warren      | 1849.840000        | 27689      | Casey (City)             | 14.970000  |
| Hadfield          | 1843.580000        | 6269       | Moreland (City)          | 3.400000   |
| Parkville         | 1829.830000        | 7074       | Melbourne (City)         | 3.870000   |
| Frankston         | 1825.020000        | 37331      | Frankston (City)         | 20.460000  |
| Altona Meadows    | 1817.840000        | 18479      | Hobsons Bay (City)       | 10.170000  |
| Ivanhoe East      | 1817.530000        | 3762       | Banyule (City)           | 2.070000   |

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Table 17: Melbourne suburbs with area and population

| Suburb             | Population Density | Population | LGA                  | Area (km2) |
|--------------------|--------------------|------------|----------------------|------------|
| Mulgrave           | 1806.880000        | 19889      | Monash (City)        | 11.010000  |
| Alphington         | 1803.340000        | 5702       | Darebin (City)       | 3.160000   |
| St Helena          | 1783.400000        | 2890       | Banyule (City)       | 1.620000   |
| Patterson Lakes    | 1771.840000        | 7793       | Kingston (City)      | 4.400000   |
| Croydon South      | 1763.890000        | 4759       | Maroondah (City)     | 2.700000   |
| Mooroolbark        | 1757.080000        | 23059      | Yarra Ranges (Shire) | 13.120000  |
| Eaglemont          | 1754.870000        | 3960       | Banyule (City)       | 2.260000   |
| Eumemmerring       | 1739.290000        | 2285       | Casey (City)         | 1.310000   |
| Williams Landing   | 1733.890000        | 9448       | Wyndham (City)       | 5.450000   |
| Ringwood           | 1726.210000        | 19144      | Maroondah (City)     | 11.090000  |
| Heidelberg West    | 1725.800000        | 5252       | Banyule (City)       | 3.040000   |
| Eltham North       | 1693.630000        | 6830       | Banyule (City)       | 4.030000   |
| Port Melbourne     | 1690.180000        | 17633      | Melbourne (City)     | 10.430000  |
| Wantirna           | 1672.880000        | 14237      | Knox (City)          | 8.510000   |
| Chelsea Heights    | 1669.270000        | 5393       | Kingston (City)      | 3.230000   |
| Bulleen            | 1669.170000        | 11219      | Manningham (City)    | 6.720000   |
| Bundoora           | 1662.880000        | 28068      | Banyule (City)       | 16.880000  |
| Kew East           | 1632.230000        | 6620       | Boroondara (City)    | 4.060000   |
| Keilor East        | 1630.420000        | 15078      | Brimbank (City)      | 9.250000   |
| Warranwood         | 1603.940000        | 4820       | Maroondah (City)     | 3.010000   |
| Clayton South      | 1592.590000        | 13381      | Kingston (City)      | 8.400000   |
| Coburg North       | 1588.840000        | 8327       | Moreland (City)      | 5.240000   |
| Sandhurst          | 1569.310000        | 5211       | Frankston (City)     | 3.320000   |
| Endeavour Hills    | 1568.470000        | 24455      | Casey (City)         | 15.590000  |
| Rowville           | 1532.710000        | 7645       | Knox (City)          | 4.990000   |
| Viewbank           | 1522.880000        | 7030       | Banyule (City)       | 4.620000   |
| Broadmeadows       | 1503.580000        | 12524      | Hume (City)          | 8.330000   |
| Bayswater          | 1483.470000        | 12262      | Knox (City)          | 8.270000   |
| Sunshine North     | 1458.130000        | 12047      | Brimbank (City)      | 8.260000   |
| Ardeer             | 1456.120000        | 3170       | Brimbank (City)      | 2.180000   |
| Strathmore Heights | 1452.970000        | 1047       | Moonee Valley (City) | 0.720000   |
| Waterways          | 1451.920000        | 2422       | Kingston (City)      | 1.670000   |
| Berwick            | 1438.260000        | 50298      | Casey (City)         | 34.970000  |
| Craigieburn        | 1420.390000        | 65178      | Hume (City)          | 45.890000  |
| Yallambie          | 1403.680000        | 4161       | Banyule (City)       | 2.960000   |
| Essendon West      | 1402.530000        | 1559       | Moonee Valley (City) | 1.110000   |
| Oakleigh South     | 1392.210000        | 9851       | Kingston (City)      | 7.080000   |
| Thomastown         | 1391.960000        | 20234      | Whittlesea (City)    | 14.540000  |
| Wantirna South     | 1376.500000        | 20754      | Knox (City)          | 15.080000  |
| Knoxfield          | 1365.420000        | 7645       | Knox (City)          | 5.600000   |
| Bayswater North    | 1337.340000        | 9014       | Maroondah (City)     | 6.740000   |
| Seaford            | 1327.980000        | 17215      | Frankston (City)     | 12.960000  |

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Table 17: Melbourne suburbs with area and population

| Suburb               | Population Density | Population | LGA                          | Area (km2) |
|----------------------|--------------------|------------|------------------------------|------------|
| Cranbourne West      | 1317.470000        | 19969      | Casey (City)                 | 15.160000  |
| Hallam               | 1315.080000        | 11355      | Casey (City)                 | 8.630000   |
| Laverton             | 1307.180000        | 4760       | Hobsons Bay (City)           | 3.640000   |
| Moorabbin            | 1286.280000        | 6287       | Kingston (City)              | 4.890000   |
| Dingley Village      | 1264.710000        | 10495      | Kingston (City)              | 8.300000   |
| Caulfield East       | 1258.140000        | 1293       | Glen Eira (City)             | 1.030000   |
| Frankston South      | 1249.930000        | 18801      | Frankston (City)             | 15.040000  |
| Kealba               | 1240.870000        | 3226       | Brimbank (City)              | 2.600000   |
| Cranbourne           | 1216.930000        | 21281      | Casey (City)                 | 17.490000  |
| Keysborough          | 1187.480000        | 30018      | Greater Dandenong (City)     | 25.280000  |
| Kilsyth              | 1167.310000        | 11699      | Maroondah (City)             | 10.020000  |
| Westmeadows          | 1166.570000        | 6502       | Hume (City)                  | 5.570000   |
| Cranbourne East      | 1164.270000        | 11177      | Casey (City)                 | 9.600000   |
| Tecoma               | 1147.110000        | 2064       | Yarra Ranges (Shire)         | 1.800000   |
| Mornington           | 1137.300000        | 25759      | Mornington Peninsula (Shire) | 22.650000  |
| Frankston North      | 1125.830000        | 5711       | Frankston (City)             | 5.070000   |
| South Morang         | 1121.470000        | 24989      | Whittlesea (City)            | 22.280000  |
| Point Cook           | 1109.730000        | 66781      | Wyndham (City)               | 60.180000  |
| Eltham               | 1104.120000        | 18847      | Banyule (City)               | 17.070000  |
| Upwey                | 1069.970000        | 6818       | Yarra Ranges (Shire)         | 6.370000   |
| Donvale              | 1056.110000        | 12644      | Manningham (City)            | 11.970000  |
| Templestowe          | 1031.020000        | 16966      | Manningham (City)            | 16.460000  |
| Carrum Downs         | 1019.840000        | 21976      | Frankston (City)             | 21.550000  |
| Upper Ferntree Gully | 1019.090000        | 3417       | Knox (City)                  | 3.350000   |
| Coolaroo             | 1017.540000        | 3193       | Hume (City)                  | 3.140000   |
| Kurunjang            | 971.630000         | 10711      | Melton (City)                | 11.020000  |
| Belgrave             | 963.700000         | 3894       | Yarra Ranges (Shire)         | 4.040000   |
| Werribee             | 940.770000         | 50027      | Wyndham (City)               | 53.180000  |
| Rosebud              | 939.680000         | 14381      | Mornington Peninsula (Shire) | 15.300000  |
| Epping               | 918.380000         | 33489      | Whittlesea (City)            | 36.470000  |
| Tarneit              | 905.020000         | 56370      | Wyndham (City)               | 62.290000  |
| Rosebud West         | 902.270000         | 5246       | Mornington Peninsula (Shire) | 5.810000   |
| Kilsyth South        | 891.140000         | 2862       | Maroondah (City)             | 3.210000   |
| Williamstown North   | 887.470000         | 1622       | Hobsons Bay (City)           | 1.830000   |
| Tullamarine          | 881.840000         | 6733       | Brimbank (City)              | 7.640000   |
| Altona North         | 871.490000         | 12962      | Hobsons Bay (City)           | 14.870000  |
| Junction Village     | 856.780000         | 1051       | Casey (City)                 | 1.230000   |
| Brookfield           | 839.190000         | 10782      | Melton (City)                | 12.850000  |
| West Melbourne       | 837.890000         | 8025       | Melbourne (City)             | 9.580000   |
| Spotswood            | 823.270000         | 2820       | Hobsons Bay (City)           | 3.430000   |
| Tootgarook           | 822.530000         | 3178       | Mornington Peninsula (Shire) | 3.860000   |
| Keilor Park          | 811.640000         | 2684       | Brimbank (City)              | 3.310000   |

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Table 17: Melbourne suburbs with area and population

| Suburb             | Population Density | Population | LGA                          | Area (km2) |
|--------------------|--------------------|------------|------------------------------|------------|
| Safety Beach       | 810.660000         | 6328       | Mornington Peninsula (Shire) | 7.810000   |
| Langwarrin         | 783.080000         | 23588      | Frankston (City)             | 30.120000  |
| Mount Eliza        | 770.770000         | 18734      | Mornington Peninsula (Shire) | 24.310000  |
| The Basin          | 767.470000         | 4497       | Knox (City)                  | 5.860000   |
| Lyndhurst          | 722.110000         | 8926       | Casey (City)                 | 12.360000  |
| Scoresby           | 692.980000         | 6066       | Knox (City)                  | 8.750000   |
| Mernda             | 690.840000         | 23369      | Whittlesea (City)            | 33.830000  |
| Attwood            | 675.830000         | 3309       | Hume (City)                  | 4.900000   |
| Melton West        | 671.540000         | 12463      | Melton (City)                | 18.560000  |
| Doreen             | 660.690000         | 27122      | Nillumbik (Shire)            | 41.050000  |
| McCrae             | 658.040000         | 3311       | Mornington Peninsula (Shire) | 5.030000   |
| Diamond Creek      | 657.830000         | 12503      | Nillumbik (Shire)            | 19.010000  |
| Altona             | 643.010000         | 11490      | Hobsons Bay (City)           | 17.870000  |
| Botanic Ridge      | 635.890000         | 6739       | Casey (City)                 | 10.600000  |
| Derrimut           | 634.610000         | 8651       | Brimbank (City)              | 13.630000  |
| Beaconsfield       | 620.920000         | 7267       | Cardinia (Shire)             | 11.700000  |
| Mount Martha       | 613.990000         | 19846      | Mornington Peninsula (Shire) | 32.320000  |
| Montrose           | 612.960000         | 6900       | Yarra Ranges (Shire)         | 11.260000  |
| Bacchus Marsh      | 583.980000         | 7808       | Moorabool (Shire)            | 13.370000  |
| Belgrave Heights   | 577.250000         | 1398       | Yarra Ranges (Shire)         | 2.420000   |
| Mount Evelyn       | 571.080000         | 9799       | Yarra Ranges (Shire)         | 17.160000  |
| Lower Plenty       | 570.360000         | 3962       | Banyule (City)               | 6.950000   |
| Lilydale           | 566.630000         | 17348      | Yarra Ranges (Shire)         | 30.620000  |
| Wyndham Vale       | 552.790000         | 12675      | Wyndham (City)               | 22.930000  |
| Keilor             | 546.140000         | 5906       | Brimbank (City)              | 10.810000  |
| Pakenham           | 543.480000         | 54118      | Cardinia (Shire)             | 99.580000  |
| Greenvale          | 506.040000         | 21274      | Hume (City)                  | 42.040000  |
| Skye               | 486.480000         | 8088       | Frankston (City)             | 16.630000  |
| Crib Point         | 484.990000         | 3343       | Mornington Peninsula (Shire) | 6.890000   |
| Rye                | 472.170000         | 9438       | Mornington Peninsula (Shire) | 19.990000  |
| Chirnside Park     | 444.640000         | 11779      | Yarra Ranges (Shire)         | 26.490000  |
| Truganina          | 442.160000         | 36305      | Melton (City)                | 82.110000  |
| Campbellfield      | 412.840000         | 4977       | Hume (City)                  | 12.060000  |
| Heatherton         | 408.460000         | 2826       | Kingston (City)              | 6.920000   |
| Park Orchards      | 403.380000         | 3835       | Manningham (City)            | 9.510000   |
| Melton             | 396.060000         | 7953       | Melton (City)                | 20.080000  |
| Melton South       | 377.720000         | 3601       | Melton (City)                | 9.530000   |
| Hastings           | 370.920000         | 10369      | Mornington Peninsula (Shire) | 27.950000  |
| Blairgowrie        | 367.960000         | 2786       | Mornington Peninsula (Shire) | 7.570000   |
| Narre Warren North | 348.470000         | 8033       | Casey (City)                 | 23.050000  |
| Brooklyn           | 339.000000         | 1979       | Brimbank (City)              | 5.840000   |
| North Warrandyte   | 337.210000         | 3027       | Nillumbik (Shire)            | 8.980000   |

Continued on next page

Table 17: Melbourne suburbs with area and population

| Suburb           | Population Density | Population | LGA                          | Area (km2) |
|------------------|--------------------|------------|------------------------------|------------|
| Ferny Creek      | 336.210000         | 1524       | Yarra Ranges (Shire)         | 4.530000   |
| Warrandyte       | 317.250000         | 5541       | Manningham (City)            | 17.470000  |
| Darley           | 310.710000         | 9190       | Moorabool (Shire)            | 29.580000  |
| Somerville       | 293.290000         | 11767      | Mornington Peninsula (Shire) | 40.120000  |
| Clyde North      | 277.570000         | 31681      | Casey (City)                 | 114.140000 |
| Research         | 275.650000         | 2695       | Nillumbik (Shire)            | 9.780000   |
| Sunbury          | 272.900000         | 38851      | Hume (City)                  | 142.360000 |
| The Patch        | 267.860000         | 1046       | Yarra Ranges (Shire)         | 3.910000   |
| Selby            | 251.910000         | 1626       | Yarra Ranges (Shire)         | 6.450000   |
| Officer          | 250.410000         | 18503      | Cardinia (Shire)             | 73.890000  |
| Hurstbridge      | 232.450000         | 3554       | Nillumbik (Shire)            | 15.290000  |
| Sorrento         | 226.970000         | 2013       | Mornington Peninsula (Shire) | 8.870000   |
| Wattle Glen      | 218.160000         | 1911       | Nillumbik (Shire)            | 8.760000   |
| Baxter           | 218.050000         | 2166       | Mornington Peninsula (Shire) | 9.930000   |
| Plenty           | 215.400000         | 2575       | Nillumbik (Shire)            | 11.950000  |
| Bittern          | 215.210000         | 4276       | Mornington Peninsula (Shire) | 19.870000  |
| Lysterfield      | 210.960000         | 6681       | Knox (City)                  | 31.670000  |
| Blind Bight      | 207.220000         | 1290       | Casey (City)                 | 6.230000   |
| Dromana          | 191.550000         | 6626       | Mornington Peninsula (Shire) | 34.590000  |
| Wandin North     | 183.530000         | 3132       | Yarra Ranges (Shire)         | 17.070000  |
| Monbulk          | 179.880000         | 3651       | Yarra Ranges (Shire)         | 20.300000  |
| Millgrove        | 176.530000         | 1666       | Yarra Ranges (Shire)         | 9.440000   |
| Badger Creek     | 174.150000         | 1610       | Yarra Ranges (Shire)         | 9.240000   |
| Wollert          | 168.110000         | 24407      | Whittlesea (City)            | 145.180000 |
| Wonga Park       | 163.500000         | 3843       | Manningham (City)            | 23.500000  |
| Wallan           | 159.090000         | 15004      | Mitchell (Shire)             | 94.310000  |
| Mount Dandenong  | 157.640000         | 1271       | Yarra Ranges (Shire)         | 8.060000   |
| Cockatoo         | 150.830000         | 4408       | Cardinia (Shire)             | 29.220000  |
| Kalorama         | 146.350000         | 1277       | Yarra Ranges (Shire)         | 8.730000   |
| Yarra Junction   | 145.510000         | 2875       | Yarra Ranges (Shire)         | 19.760000  |
| Langwarrin South | 142.220000         | 1346       | Frankston (City)             | 9.460000   |
| Tyabb            | 141.340000         | 3449       | Mornington Peninsula (Shire) | 24.400000  |
| Maddingley       | 135.080000         | 5491       | Moorabool (Shire)            | 40.650000  |
| Belgrave South   | 129.550000         | 1670       | Yarra Ranges (Shire)         | 12.890000  |
| Kallista         | 119.220000         | 1418       | Yarra Ranges (Shire)         | 11.890000  |
| Emerald          | 114.590000         | 5890       | Cardinia (Shire)             | 51.400000  |
| Pearcedale       | 113.460000         | 3867       | Casey (City)                 | 34.080000  |
| Somers           | 111.800000         | 1857       | Mornington Peninsula (Shire) | 16.610000  |
| Seville          | 111.240000         | 2559       | Yarra Ranges (Shire)         | 23.000000  |
| Woori Yallock    | 107.680000         | 2964       | Yarra Ranges (Shire)         | 27.530000  |
| Yarrambat        | 103.490000         | 1602       | Nillumbik (Shire)            | 15.480000  |
| Bunyip           | 101.830000         | 3131       | Cardinia (Shire)             | 30.750000  |

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Table 17: Melbourne suburbs with area and population

| Suburb             | Population Density | Population | LGA                          | Area (km2) |
|--------------------|--------------------|------------|------------------------------|------------|
| Beaconsfield Upper | 100.400000         | 2997       | Cardinia (Shire)             | 29.850000  |
| Olinda             | 97.900000          | 1773       | Yarra Ranges (Shire)         | 18.110000  |
| Gisborne           | 93.100000          | 10142      | Macedon Ranges (Shire)       | 108.940000 |
| New Gisborne       | 91.020000          | 2509       | Macedon Ranges (Shire)       | 27.570000  |
| Ravenhall          | 84.760000          | 2295       | Melton (City)                | 27.080000  |
| Cranbourne South   | 83.710000          | 3241       | Casey (City)                 | 38.720000  |
| Balnarring         | 82.660000          | 2371       | Mornington Peninsula (Shire) | 28.680000  |
| Yarra Glen         | 81.560000          | 3012       | Yarra Ranges (Shire)         | 36.930000  |
| Garfield           | 67.600000          | 2114       | Cardinia (Shire)             | 31.270000  |
| Harkaway           | 67.530000          | 1011       | Casey (City)                 | 14.970000  |
| Hmas Cerberus      | 67.520000          | 1124       | Mornington Peninsula (Shire) | 16.650000  |
| Launching Place    | 66.280000          | 2495       | Yarra Ranges (Shire)         | 37.640000  |
| Devon Meadows      | 65.540000          | 1551       | Casey (City)                 | 23.660000  |
| Koo Wee Rup        | 60.680000          | 4047       | Cardinia (Shire)             | 66.690000  |
| Werribee South     | 58.490000          | 2392       | Wyndham (City)               | 40.900000  |
| Romsey             | 58.430000          | 5797       | Macedon Ranges (Shire)       | 99.210000  |
| Mickleham          | 57.410000          | 17452      | Hume (City)                  | 303.990000 |
| Tooradin           | 56.550000          | 1722       | Cardinia (Shire)             | 30.450000  |
| Riddells Creek     | 54.370000          | 4390       | Macedon Ranges (Shire)       | 80.740000  |
| Clyde              | 53.650000          | 11177      | Casey (City)                 | 208.330000 |
| Whittlesea         | 52.810000          | 6117       | Whittlesea (City)            | 115.830000 |
| Healesville        | 52.640000          | 7589       | Yarra Ranges (Shire)         | 144.170000 |
| Macedon            | 52.340000          | 2073       | Macedon Ranges (Shire)       | 39.610000  |
| Panton Hill        | 50.360000          | 1063       | Nillumbik (Shire)            | 21.110000  |
| Eden Park          | 49.230000          | 1194       | Whittlesea (City)            | 24.250000  |
| Eynesbury          | 42.290000          | 2838       | Melton (City)                | 67.110000  |
| Silvan             | 41.290000          | 1323       | Yarra Ranges (Shire)         | 32.040000  |
| Diggers Rest       | 40.780000          | 5669       | Hume (City)                  | 139.010000 |
| Wandong            | 39.560000          | 1477       | Mitchell (Shire)             | 37.340000  |
| Mount Macedon      | 39.250000          | 1450       | Macedon Ranges (Shire)       | 36.940000  |
| Red Hill           | 39.170000          | 1009       | Mornington Peninsula (Shire) | 25.760000  |
| Coldstream         | 38.910000          | 2199       | Yarra Ranges (Shire)         | 56.520000  |
| Pakenham Upper     | 36.610000          | 1196       | Cardinia (Shire)             | 32.670000  |
| Rockbank           | 36.570000          | 7982       | Melton (City)                | 218.270000 |
| Kangaroo Ground    | 34.850000          | 1208       | Nillumbik (Shire)            | 34.660000  |
| St Andrews         | 30.810000          | 1186       | Nillumbik (Shire)            | 38.490000  |
| Moorooduc          | 25.610000          | 1004       | Mornington Peninsula (Shire) | 39.200000  |
| Wesburn            | 24.690000          | 1052       | Yarra Ranges (Shire)         | 42.610000  |
| Beveridge          | 20.860000          | 4642       | Whittlesea (City)            | 222.530000 |
| Lancefield         | 19.830000          | 2743       | Macedon Ranges (Shire)       | 138.330000 |
| Nar Nar Goon       | 19.470000          | 1023       | Cardinia (Shire)             | 52.540000  |
| Kinglake West      | 17.860000          | 1305       | Murrindindi (Shire)          | 73.070000  |

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*Table 17: Melbourne suburbs with area and population*

| Suburb    | Population Density | Population | LGA                          | Area (km <sup>2</sup> ) |
|-----------|--------------------|------------|------------------------------|-------------------------|
| Flinders  | 16.420000          | 1130       | Mornington Peninsula (Shire) | 68.820000               |
| Warburton | 15.180000          | 2020       | Yarra Ranges (Shire)         | 133.070000              |
| Kalkallo  | 14.950000          | 5548       | Hume (City)                  | 371.100000              |
| Kinglake  | 13.340000          | 1662       | Murrindindi (Shire)          | 124.590000              |
| Gembrook  | 11.610000          | 2559       | Cardinia (Shire)             | 220.410000              |

## APPENDIX 2 DESIGN CRITERIA FOR PRIMARY SEDIMENTATION BASINS

Table 18: Typical design criteria for primary sedimentation basins [Davis, 2010]

| Parameter                     | Range of values                            | Typical/comment                                     |
|-------------------------------|--------------------------------------------|-----------------------------------------------------|
| <i>General</i>                |                                            |                                                     |
| Overflow rate (average flow)  | 30 to 50 m <sup>3</sup> /d·m <sup>2</sup>  | 40 m <sup>3</sup> /d·m <sup>2</sup>                 |
| Overflow rate (peak flow)     | 60 to 120 m <sup>3</sup> /d·m <sup>2</sup> | 100 m <sup>3</sup> /d·m <sup>2</sup>                |
| Detention time (average flow) | 1.5 to 2.5 h                               | 2.0 h                                               |
| Flow velocity                 | 0.020 to 0.025 m/s                         | –                                                   |
| Weir loading rate             | 125 to 500 m <sup>3</sup> /d·m             | 250 m <sup>3</sup> /d·m                             |
| Sludge hoppers                | 1.7 vertical to 1 horizontal               | Minimum; bottom width < 0.6 m                       |
| Geotechnical                  | –                                          | Consider potential for flotation when tank is empty |
| <i>Circular tanks</i>         |                                            |                                                     |
| <i>Dimensions</i>             |                                            |                                                     |
| Diameter                      | 3 to 100 m                                 | 12 to 45 m                                          |
| Standard                      | 9 to 45 m                                  | In 1.5 m increments                                 |
| Side water depth              | 3 to 5 m                                   | 4.3 m                                               |
| Floor slope                   | 1 vertical to 12 horizontal                | –                                                   |
| Splitter box inlet velocity   | < 0.3 m/s                                  | At peak flow                                        |
| <i>Inlet configuration</i>    |                                            |                                                     |
| Detention time                | 20 minutes                                 | Feedwell                                            |
| Submergence                   | 30 to 75% of depth                         | Size to prevent scour                               |
| EDI detention time            | 8 to 10 s                                  | –                                                   |